Reg. No. | | | | | | | | | | | [18MTDS243]

SECOND SEMESTER M.TECH EXAMINATION, JUNE 2019

PATTERN RECOGNITION

Time: 03 Hours **Maximum Marks:100**

Instructions: **i. Answer FIVE full questions. Question No. 1 and 2 are compulsory.**
ii. Draw neat sketches wherever necessary.
iii. Make suitable assumptions wherever necessary.

1.	**a.**	What is feature extraction? Explain with the help of an example.	**10**
	b.	Explain the basic concept of pattern recognition.	**10**
2.	**a.**	Describe Fisher's linear discriminant analysis for two class in detail.	**10**
	b.	What is Gaussian distribution? Explain with the help of an example.	**10**
3.	**a.**	Describe dimension reduction in pattern recognition? Also explain different methods for dimension reduction.	**10**
	b.	Explain the maximum likelihood method for parameter estimation.	**10**
		OR	
4	**a.**	Discuss stopping criteria in decision tree. Also describe why pruning is an important step while creating decision tree.	**10**
	b.	Explain Gaussian mixture models. Also discuss EM algorithm.	**10**
5.	**a.**	Discuss examples of kernels commonly used in SVM learning.	**10**
	b.	How Tic-Tac-Toe can be solved using Reinforcement learning?	**10**
		OR	
6.	**a.**	Write short notes on :	**10**
		i. Relaxation procedures	
		ii. Minimum error rate classifier	
	b.	Discuss Ho0kashapy procedure in detail.	**10**
7.	**a.**	What is markou model? Describe first-order hidden markou models.	**10**
	b.	How CART can be used for classification with an example.	**10**
		OR	
8.	**a.**	Describe Bayes decision rule. Describe the input necessary to arrive at a Bayes decision rule for a given data set.	**10**
	b.	Explain Boltzmann learning in detail.	**10**

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